FARIA · 7 NOVEMBER 2026 · A LESSON FOR JEFF FRANZEN

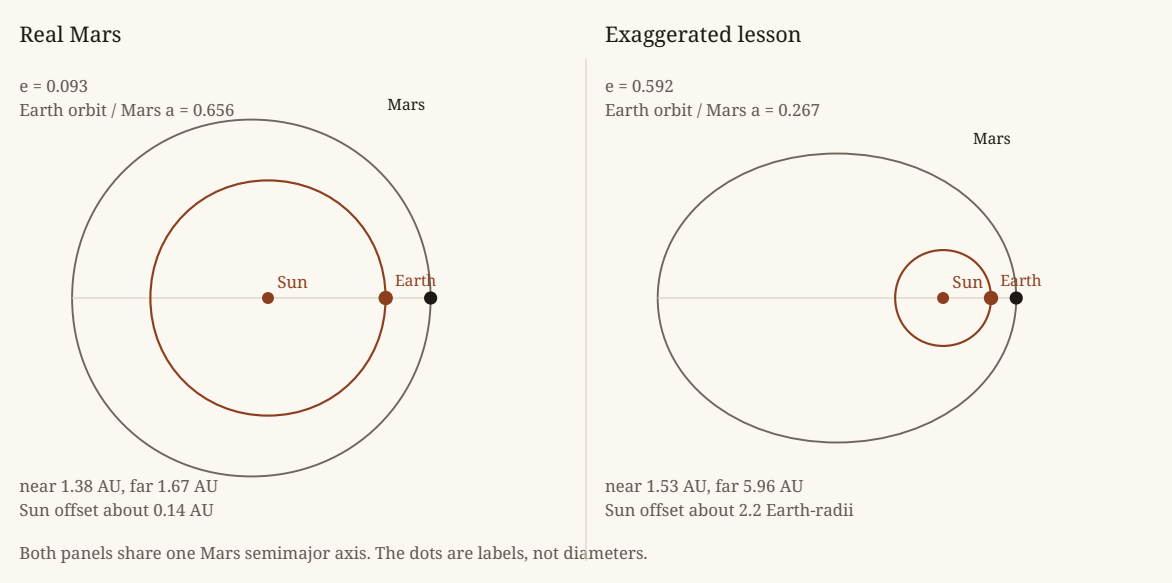
Kepler's Mars triangles, and an honest scale check

Kepler did not guess Mars's orbit from one night. Mars returns to the same place in its path about every 687 days. Earth has moved. If you know where Earth is, two lines of sight to that same Martian place cross, and you have a point in space. Angles measured against the stars give the directions. One angle plus a guessed radius is not enough. Kepler also had to repair Earth's own orbit before the crossings would sit still. Many pairs, not one clever triangle, drew the curve. It is an ellipse, with the Sun at a focus.

Opposition and conjunction are easy to mix up with a measurement. At opposition the order is Sun, Earth, Mars, with Earth in the middle. Mars is up all night and near its brightest. At conjunction the order is Earth, Sun, Mars. Mars is lost in the glare. Nothing is eclipsed. A straight alignment is not, by itself, a distance. Hipparchus's lesson uses a short baseline on the Earth. Here the baseline is Earth's whole orbit, the same stick that, once it has a length in miles, carries the transit lesson outward.

An early illustrated sheet of this ellipse, made to be obvious in a small picture, stretched the truth and said so in a caption. It was not a printer's accident. On that sheet Mars's semiaxes were 247 and 199 drawing units, and Earth's orbital radius was 66. The ratio is about 3.74, against the real ratio of semimajor axes, about 1.524. The drawing is about 2.46 times too stretched. Its eccentricity is about 0.59, against Mars's real 0.093, roughly six times too oval. The Sun sits about 2.2 of those Earth-radii from the center. In the sky the offset is about 0.14 AU. If you treat the drawn Earth radius as 1 AU, the sheet's nearest and farthest distances are about 1.53 and 5.96 AU. The real ones are about 1.38 and 1.67 AU.

Earth's orbital radius ought to be about 66 percent of Mars's semimajor axis, not about 27 percent. That is why Earth looks tiny on the old panel. The tininess is the exaggeration. It is the wrong reason for the eye to think Earth is small. The planet dots on that sheet were 14 and 12 units across, a ratio near 1.17. Mars's diameter is about 6,792 kilometers and Earth's about 12,756, a ratio near 0.53. Dots are names, not worlds.



Real Mars beside the exaggerated teaching ellipse

Both panels use the same Mars semimajor axis, and both put the Sun at the focus. The right-hand ellipse is the exaggerated lesson, labeled as such. The dots are not the planets' physical sizes.

The illustrated lesson is download the PDF.

Sources

- Steven Austin, notes on Kepler's Mars orbit: https://faculty.uca.edu/saustin/3110/mars.pdf
- Physics Today, on Kepler and Mars: https://physicstoday.aip.org/features/the-great-martian-catastrophe-and-how-kepler-fixed-it
- NASA Mars fact sheet: https://nssdc.gsfc.nasa.gov/planetary/factsheet/marsfact.html
- NASA, Mars at opposition: https://science.nasa.gov/asset/hubble/close-encounter-mars-at-opposition/
- NASA, geometry of a Mars conjunction: https://science.nasa.gov/resource/geometry-of-mars-solar-conjunction/

Key facts

- A repeated heliocentric place of Mars, seen from two known positions of Earth, fixes a point. Many points make the ellipse. One angle is not enough.
- Opposition puts Earth between the Sun and Mars. Conjunction hides Mars in the Sun's glare. Neither alignment is a distance by itself.
- The exaggerated sheet has orbital ratio about 3.74 instead of 1.52, and eccentricity about 0.59 instead of 0.093. Earth's orbit should be about 66 percent of Mars's semimajor axis, not 27 percent. The dots are not sizes: Mars is about half Earth's diameter.